

Seasonal influenza circulation patterns and projections for September 2026 to September 2027

John Huddleston¹, Jover Lee¹, Philippa Steinberg¹, Caroline Kikawa², Andrew Butler²,
Jesse D. Bloom^{2,3}, Trevor Bedford¹ & Richard A. Neher⁴

¹Vaccine and Infectious Disease Division, Fred Hutchinson Cancer Center, Seattle, WA, USA, ²Division of Basic Sciences and Computational Biology Program, Fred Hutchinson Cancer Center, Seattle, WA, USA, ³Howard Hughes Medical Institute, Seattle, WA, USA, ⁴Biozentrum, University of Basel, Basel, Switzerland

September 18, 2026

Abstract

This report details current seasonal influenza circulation patterns as of early September 2026 and was prepared for the Southern Hemisphere VCM on September 18, 2026. This is not meant as a comprehensive report, but is instead intended as particular observations that we’ve made that may be of relevance. Please also note that observed patterns reflect the GISAID database and may not be entirely representative of underlying dynamics. All analyses are based on the Nextstrain pipeline [1,2] with continual updates posted to nextstrain.org/seasonal-flu. *In compliance with data sharing agreements, this public version of the report does not include raw serological measurements.*

A/H1N1pdm: HA clade D.3.1.1 is now the dominant clade globally. D.3.1.1 subclades with HA1 139N and 155E are growing rapidly and show strong antigenic advance in human and ferret serology. Based on human serology, we predicted that vaccine candidates from D.3.1.1 with 139N and 155E (A/Tokyo/EIS15-043/2026 and A/Sydney/81/2026) were antigenically closest to predicted future H1N1pdm populations.

A/H3N2: HA clade K has fixed globally. Multiple emerging variants within K appear to have higher growth rates than ancestral K viruses. Variants of K with HA1 223I or 79I/209Q show the most antigenic advance in human serology. A K:328T variant reflects a high fitness reversion to a pre-K allele with no clear antigenic effect. From human serology, we predicted that the current Northern Hemisphere vaccine, A/Darwin/1415/2025, was among the closest strains to predicted future H3N2 populations.

B/Vic: HA diversity has persisted with C.3 and C.5 descendants co-circulating and diversifying. However, C.3.1, C.3.3, C.5.1, and C.5.7 all appear to be declining in frequency, while C.5.6 variants like C.5.6.1 with HA1 241Q are growing. C.3 descendants with HA1 197N have the highest antigenic advance in ferrets followed by C.5.6.1 variants with 241Q.

Contents

Methods	3
A/H1N1pdm	6
Summary	6
Antigenic properties from ferret serology	9
Antigenic properties from human serology	10
Comparing viral fitness forecasts and antigenic phenotype	11
A/H3N2	13

Summary	13
Antigenic properties from ferret serology	15
Antigenic properties from human serology	16
Comparing viral fitness forecasts and antigenic phenotype	18
B/Vic	20
Summary	20
Antigenic properties from ferret serology	22

Methods and Notes

Sequence data and subsampling

We base our analysis on sequence data available in the GISAID EpiFlu Database [3] as of September 16, 2026 and titer data available as of September 17, 2026. The availability of sequences varies greatly across time and geography and we try to minimize geographical and temporal bias by subsampling the data or analyzing different geographical regions separately when appropriate. While this subsampling reduces geographical biases, it doesn't remove this bias entirely.

Phylogenetic analysis

We build geographically-representative phylogenies to show the relationships among major genetic variants. To minimize bias from differences in surveillance intensity, we evenly sample approximately 3,000 sequences collected in the last 2 years across major continental regions. We sample 2,700 sequences by region, year, and month for all regions except Europe. From Europe, we sample 300 sequences by country, year, and month.

For all other phylogenetic analyses, we use all available sequences collected since February 1, 2026 and include all titer reference strains from the last eight years, to ensure that all titer data are available in our interactive reports. We build these phylogenies with IQ-TREE's implementation of the MAPLE maximum parsimony algorithm [4].

Repeated occurrence of identical substitutions in different clades of the tree is common in A/H3N2 and A/H1N1pdm. Such parallel evolution violates fundamental assumptions of common phylogeny software and can erroneously group distinct clades together if they share too many parallel changes. To avoid such artifacts, we mask sites with rampant parallelism prior to phylogeny inference.

We assign clades using a collection of "signature" mutations available at the GitHub repository github.com/nextstrain/seasonal-flu for each lineage via the following links:

- H1N1pdm HA
- H1N1pdm NA
- H3N2 HA
- H3N2 NA
- Vic HA
- Vic NA

Frequency and fitness estimates

We apply a non-phylogenetic method to estimate clade frequencies from all hemagglutinin (HA) sequences per lineage. We align all HA sequences to the lineage's reference sequence using Nextclade and obtain clade labels and amino acid substitutions per sequence [5]. From these Nextclade annotations per sequence, we jointly estimate the frequencies and relative growth advantages of each clade per geographic region in 7-day intervals using a hierarchical multinomial logistic regression (MLR) model [6]. Multinomial logistic regression across n variants models the probability of a virus sampled at time t

belonging to variant i as equal to its frequency $x_i(t)$ following

$$\Pr(X = i) = x_i(t) = \frac{p_i \exp(f_i t)}{\sum_j p_j \exp(f_j t)}.$$

A toy example of MLR fit to frequency data is shown in Figure 1.

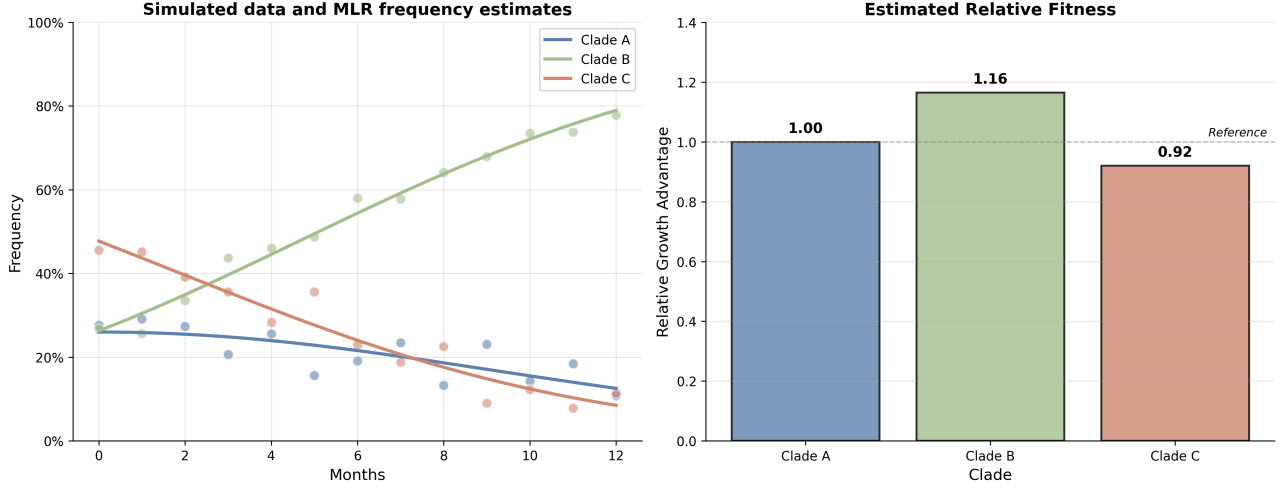


Figure 1. Demo of MLR model and directly estimating clade-level growth advantage from frequency dynamics.

Growth advantages reflect the advantage of each clade relative to a specific reference clade or “pivot” clade. Values less than 1 indicate lower fitness than the pivot clade and values greater than 1 indicate higher fitness. In addition to estimating region-specific growth advantages, the model estimates an average growth advantage and variance per clade across all regions which we report as the “hierarchical GA”. The hierarchical GA is analogous to the local branching index in that it reflects the recent global success of each clade. The region-specific GA is analogous to a region-specific local branching index.

Antigenic analysis of ferret data

We summarize HI and FRA measurements provided by the WHO CCs in London, Melbourne, Atlanta, and Tokyo using our *substitution model* [7] which models log-titers as a sum of effects associated with amino acid differences between the sequences of the test and reference virus. In addition, the model allows for a serum (column) and a virus (row) effect. This model allows inference of titers for virus/serum pairs that have not been antigenically characterized and isolates effects consistently observed across many measurements from the noise inherent in individual measurements. We also plot individual and average normalized $\log_2$ titers per reference serum for test viruses in specific clades, to represent the raw antigenic data available for each influenza lineage.

In addition to the MLR-based growth advantages described above, we estimated the fitness of currently circulating strains using antigenic advance based on titer measurements from ferret sera [7]. High antigenic advance indicates groups of viruses with amino acid substitutions that are inferred to increase the antigenic distance from viruses without those substitutions.

High throughput human serology and antigenic analysis

Kikawa, Butler et al. [8] describes the generation of a library of recently circulating H3N2 and H1N1pdm strains. This work uses a high-throughput sequencing-based neutralization assay to measure neutralization titers for these 148 viruses against a panel of 325 human sera collected from individuals spanning from young children to elderly adults, drawn from multiple sites around the world and collected between April and August 2026 (Fig. 2). We apply a similar ‘titer model’ as used for ferret data [7] to predict antigenic escape for strains that were not directly measured in the assay. In this report, we focus on neutralization titers for a subset of sera (N=244) that excludes post-vaccination serum samples.

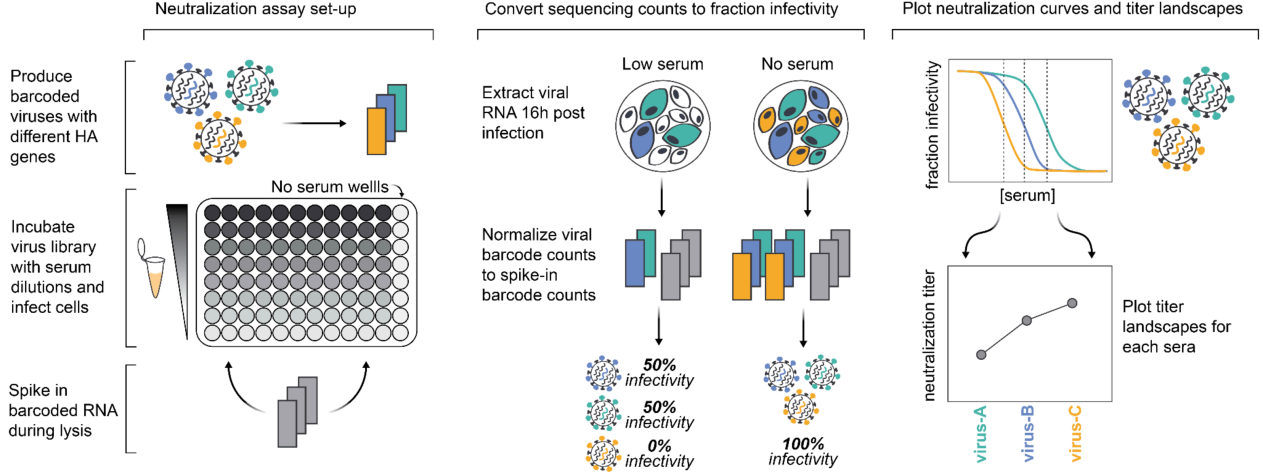


Figure 2. High-throughput sequencing-based neutralization assays enable rapid measurement of human serological responses across many viruses. Figure reproduced from Kikawa et al. [9].

Antigenic distance to the predicted future population

To rank vaccine candidates per subtype by their antigenic distance to the future, we integrated antigenic distance estimated from human or ferret serological data with predicted future frequencies estimated by the MLR model. The MLR model allows us to predict future frequencies of emerging haplotypes per geographic region. Our titer substitution model [7] estimates the antigenic effect of each HA1 substitution between reference and test viruses. For each reference virus used in ferret cell HI experiments (vaccine candidates), we calculated the average antigenic distance between that candidate’s HA1 amino acid sequence and the corresponding HA1 sequence for each virus in each combination of emerging haplotype and geographic region. We calculated each pairwise antigenic distance between the candidate and another virus sequence as the sum of antigenic effects associated with the substitutions between the two viral HA1 sequences. We calculated a weighted average antigenic distance to the future by multiplying the predicted future frequency of each haplotype in a given region (e.g., Fig. 4) by the average antigenic distance for that haplotype/region combination. Finally, we averaged these estimates across regions to get the weighted average antigenic distance for the vaccine strain to the predicted future population. Candidates with the lowest distances are predicted to be the best representatives of the future population.

A/H1N1pdm

Summary

HA clade D.3.1.1 is now the dominant clade globally (Figs. 3 and 4). D.3.1.1 subclades with 139N and 155E are growing rapidly everywhere (Fig. 4). One subclade with 139N, 155E, and 152I is the most geographically distributed (Fig. 5), while the other subclade with 139N, 155E, and 54H appears primarily in Southeast Asia. This rapid growth of these subclades indicates their high fitness relative to all other variants (Fig. 6).

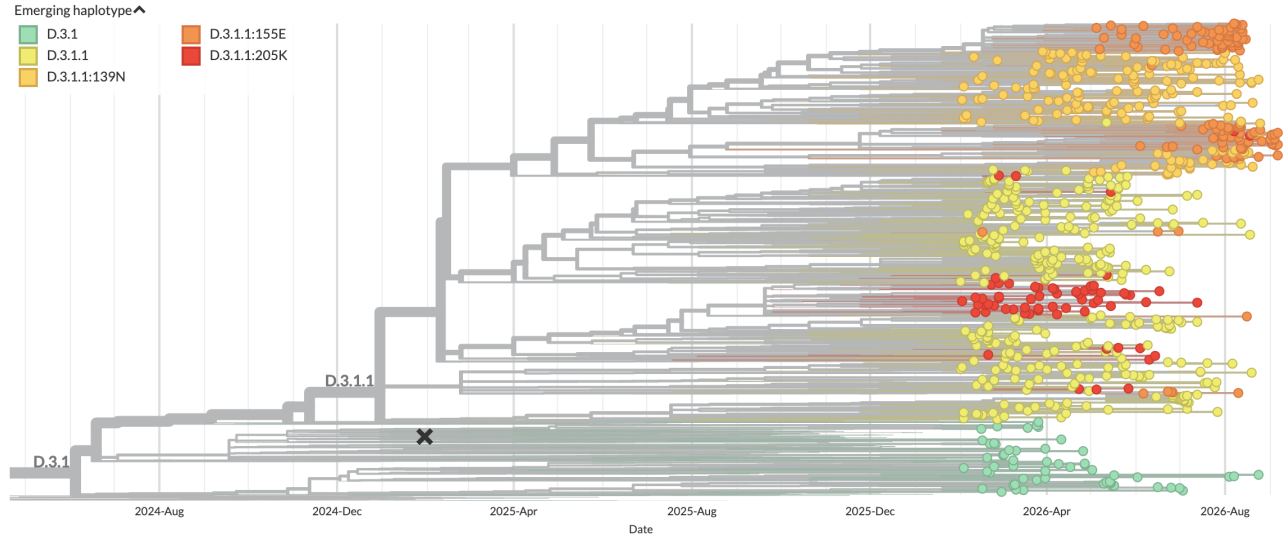


Figure 3. Time-resolved A/H1N1pdm phylogeny colored by clade and filtered to strains collected since February 1, 2026. View on nextstrain.org.

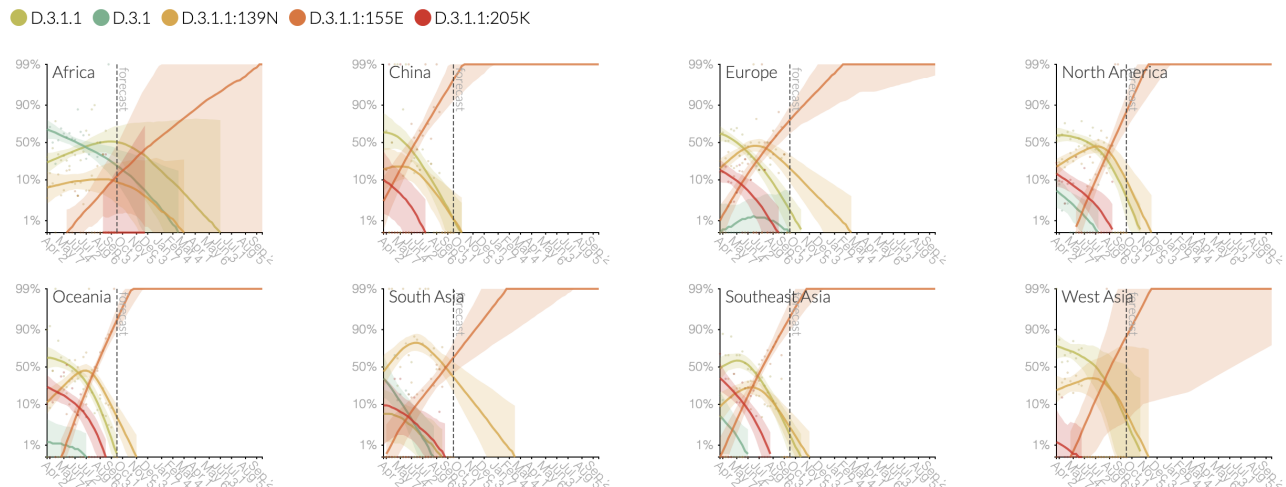


Figure 4. Clade frequencies by region estimated by a multinomial logistic regression (MLR) model. Lines show the median frequency ($\pm$ 95% HPDIs) and 1-year forecasts estimated under an assumption of exponential growth. Colored points indicate frequency estimates from the raw data binned into weekly intervals. View on [Nextstrain](https://nextstrain.org).

○ D.3.1.1 ○ D.3.1 ○ D.3.1.1:D139N ● D.3.1.1:D139N-V152I-G155E ● D.3.1.1:Q54H-D139N-G155E
○ D.3.1.1:R205K

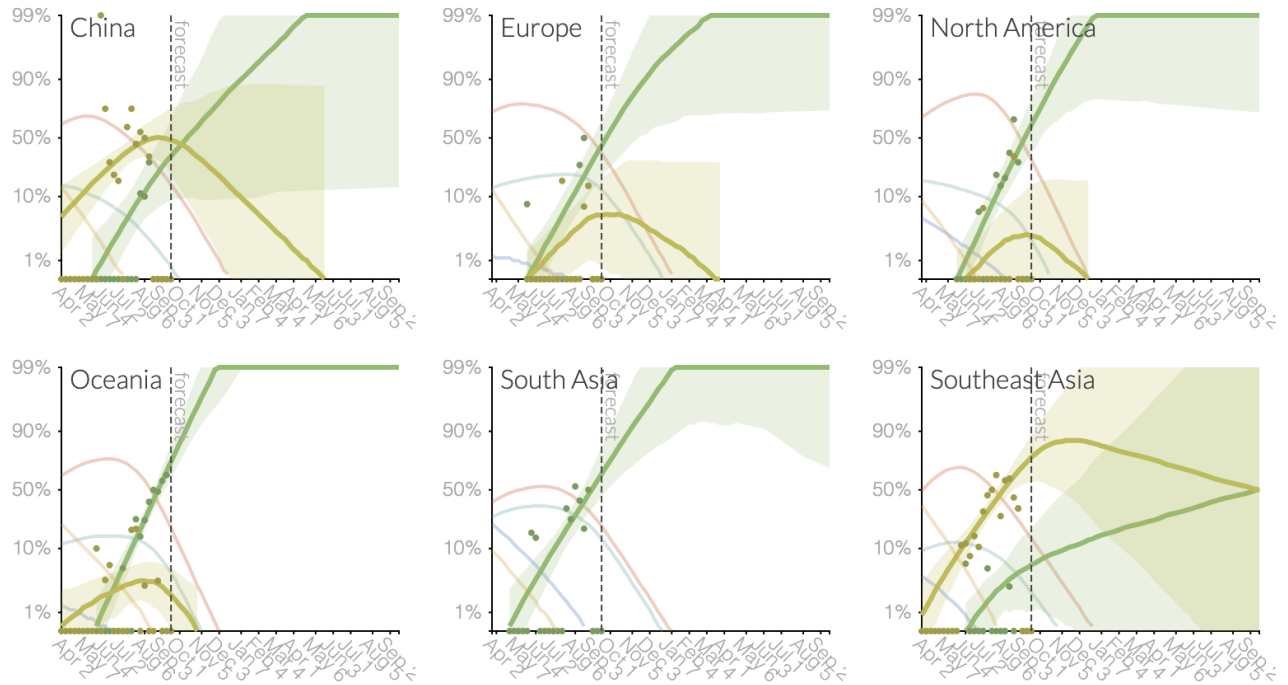


Figure 5. Amino acid haplotype frequencies by region estimated by a multinomial logistic regression (MLR) model. Lines show the median frequency ($\pm$ 95% HPDIs) and 1-year forecasts estimated under an assumption of exponential growth. Colored points indicate frequency estimates from the raw data binned into weekly intervals. View on Nextstrain.

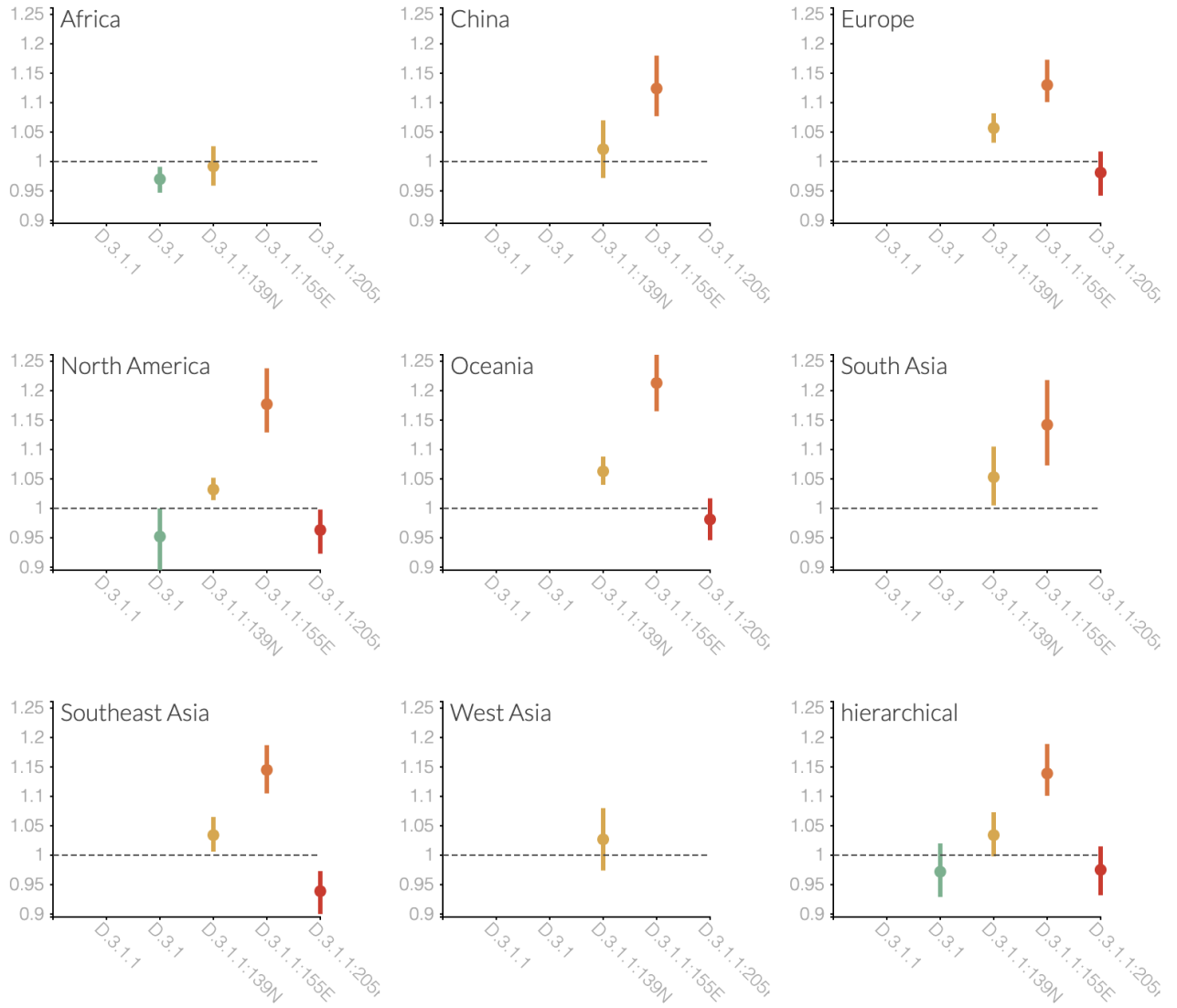


Figure 6. Growth advantages per clade by region as estimated by a MLR model. The “hierarchical” advantage represents the global average fitness per clade. The dashed vertical line at $x=1$ indicates the growth advantage of the pivot clade (D.3.1.1). Clades with growth advantages greater than 1 have a higher fitness than the pivot. Colored points indicate the median growth advantage per clade and location ($\pm$ 95% HPDIs). View on Nextstrain.

Antigenic properties from ferret serology

D.3.1.1 viruses with G155E substitutions continue to show the highest antigenic advance across all genetic backgrounds with an effect of 2.0 $\log_2$ units (Fig. 7). The HA1:V152I substitution increases antigenic advance by 0.4 $\log_2$ units.

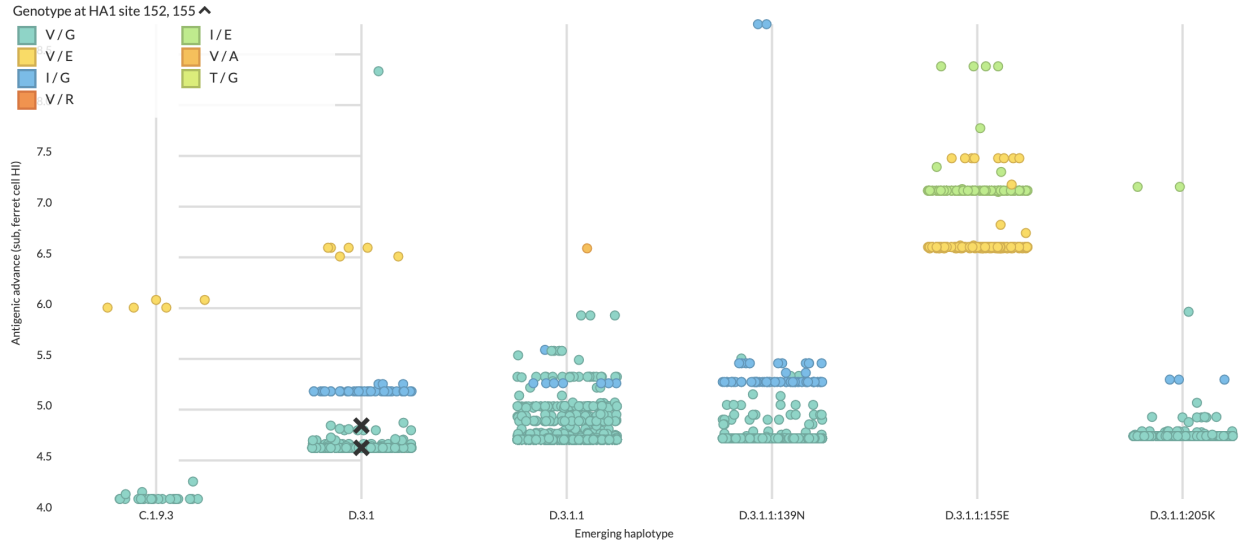


Figure 7. Antigenic advance across clades estimated from ferret-based HI data [7] for samples collected since February 1, 2026. View on nextstrain.org.

[Raw ferret data redacted]

Antigenic properties from human serology

The Bloom Lab measured neutralization titers for recent H3N2 and H1N1pdm strains against human sera collected in April through August 2026 (see methods summary at the beginning of the report). Results and data are available in a preprint posted on bioRxiv [8] and summarized in the figures below (Fig. 8).

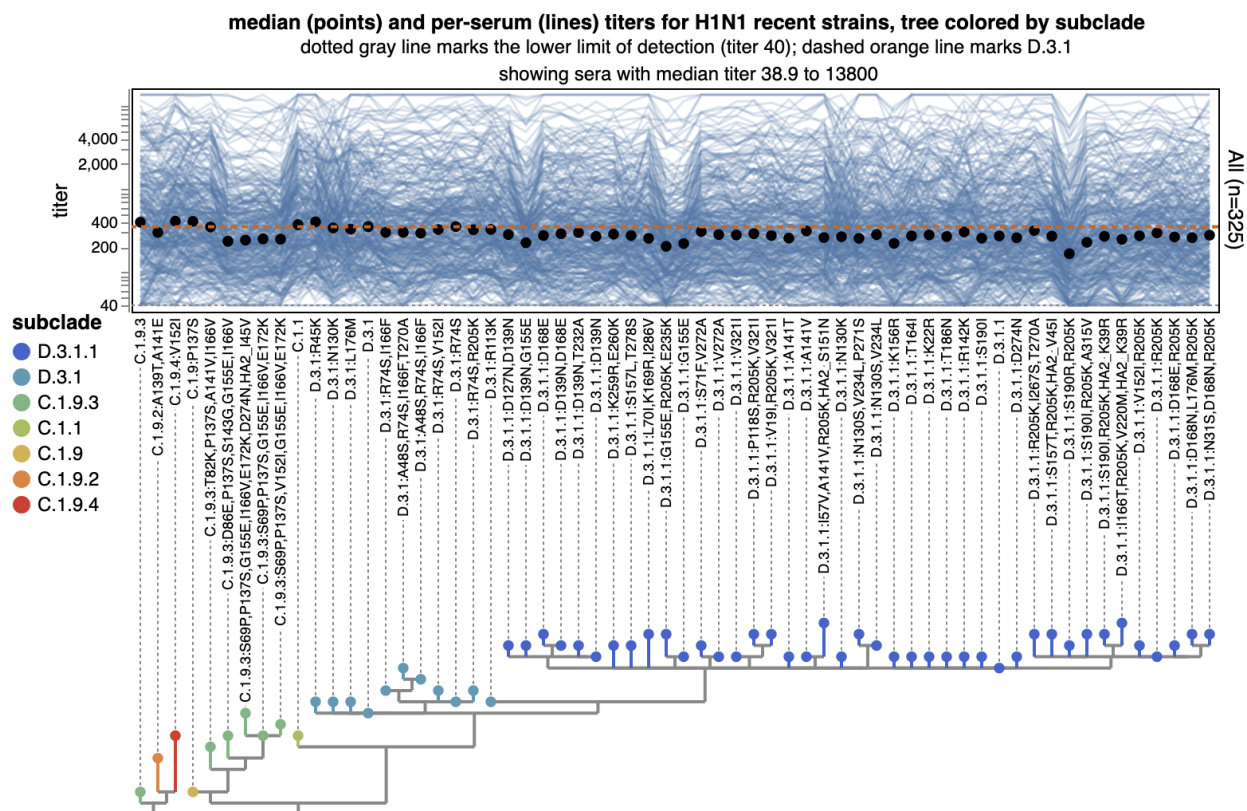


Figure 8. Tree summarizing H1N1pdm strains used by Kikawa, Butler et al. [8] and individual titers per virus and serum. The solid points represent the median titer across all sera. The dashed line represents the titer to the current H1N1pdm vaccine strain.

Comparing viral fitness forecasts and antigenic phenotype

To understand the relationship between viral fitness estimated from frequency dynamics (growth advantages from the MLR model) and antigenic advance estimated from neutralization assays, we plotted the growth advantage of each recent emerging haplotype against the corresponding median antigenic advance of sequences in that haplotype based on human or ferret serological data. Human antigenic advance corresponds with estimated fitness in nature better than ferret antigenic advance does (Fig. 9). For example, greater antigenic advance for D.3.1.1 and its subclades with 139N and 155E compared to D.3.1 correspond linearly with higher fitness. In contrast, we see no distinction between most recent clades by antigenic advance from ferrets except for D.3.1.1:155E.

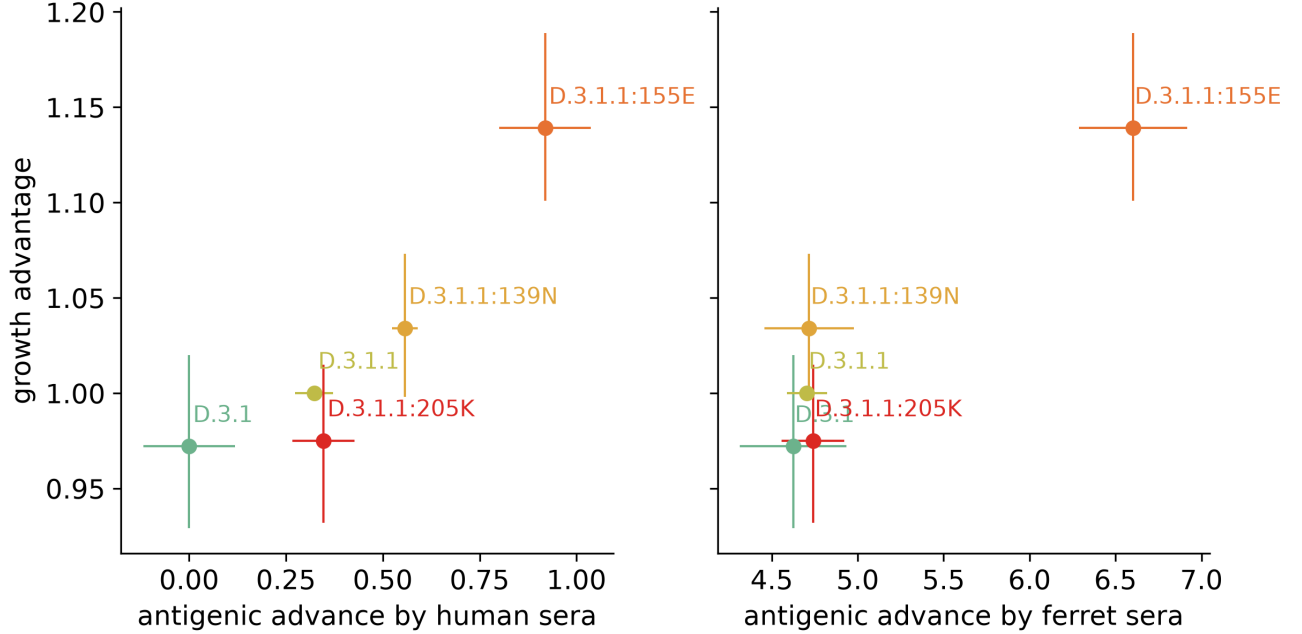


Figure 9. Growth advantage by antigenic advance per clade estimated with a titer substitution model [7] for human-based neutralization data [10] (left) and ferret-based cell HI data (right). Growth advantages shown as median $\pm$ 95% HPDIs. Antigenic advance shown as median $\pm$ 1 standard deviation across all sequences in a given clade.

Finally, we ranked the available vaccine candidates by their antigenic distance to the predicted future H1N1pdm population as described in the Methods. We found that reference viruses from D.3.1.1 with 139N and 155E (A/Tokyo/EIS15-043/2026 and A/Sydney/81/2026) were closest to the predicted future (0.09 $\log_2$ units away), followed by D.3.1.1 viruses with only 155E (A/Tokyo/EIS13-961/2025, 0.26 $\log_2$ units) or 139N (A/Massachusetts/22/2026, 0.44 $\log_2$ units, Fig. 10). In contrast, ferret-based antigenic distances to the future ranked all D.3.1.1 candidates with 155E similarly well (Fig. 11).

A/H3N2

Summary

HA clade K has fixed globally (Figs. 12 and 13). Multiple emerging haplotypes within K appear to be more fit than ancestral K viruses including K:96R, K:328T, K:223I, and K:79I-207Q (Fig. 14). Of these, only K:79I-207Q is monophyletic. Different regions have different rapidly growing haplotypes: K:223I in Oceania, K:328T in Europe, K:79I-207Q in Southeast Asia and China (Fig. 13). As a result, we cannot confidently predict which haplotype will outcompete the others in the next year.

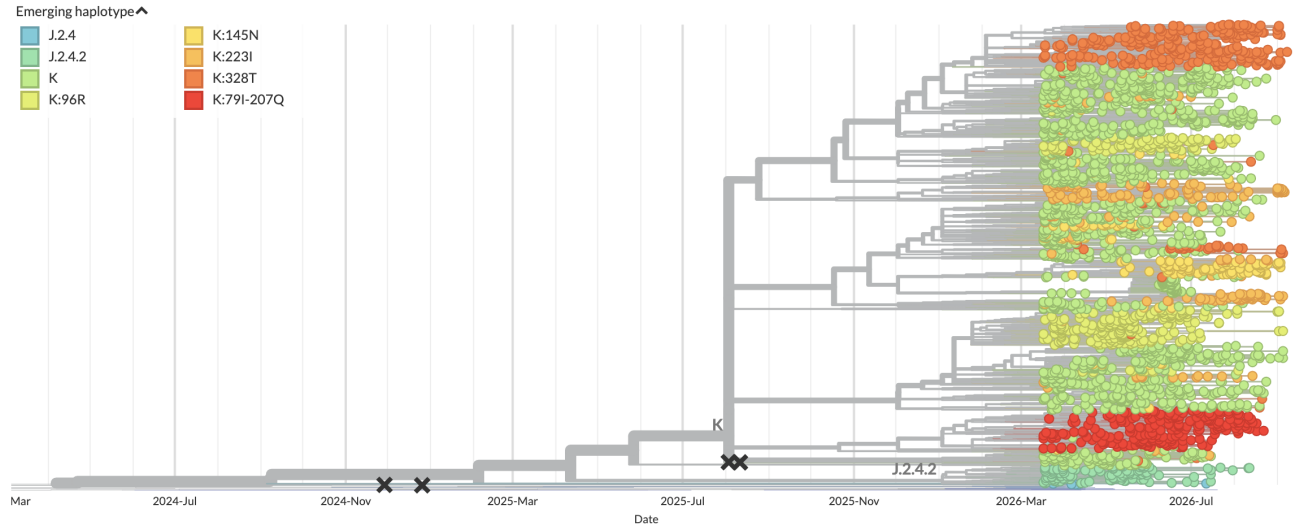


Figure 12. Time-resolved A/H3N2 phylogeny colored by clade and filtered to strains collected since March 2026 View on nextstrain.org.

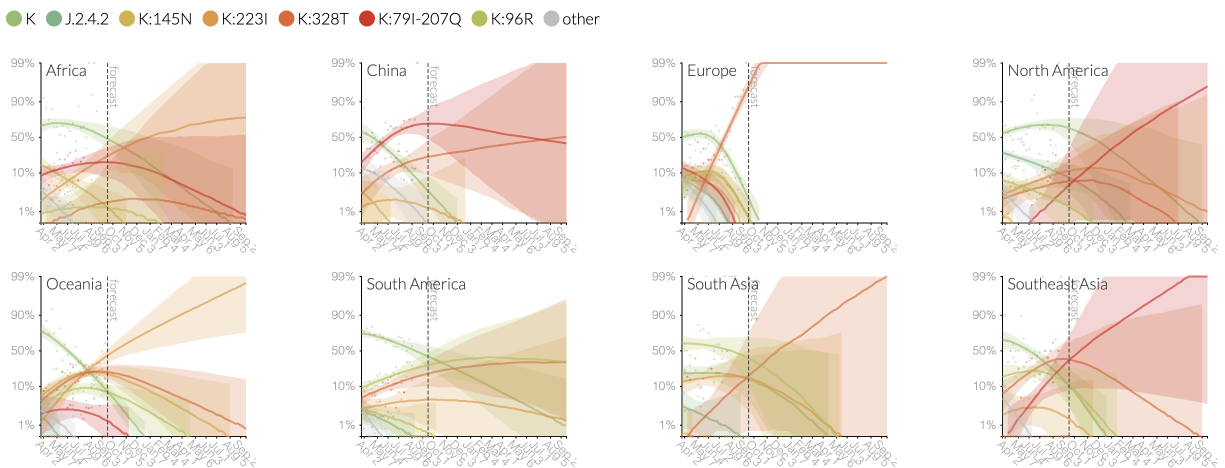


Figure 13. Clade frequencies by region estimated by a multinomial logistic regression (MLR). Lines show the median frequency ($\pm$ 95% HPDIs) and 1-year forecasts. Colored points indicate frequency estimates from the raw data. View on [Nextstrain](https://nextstrain.org).

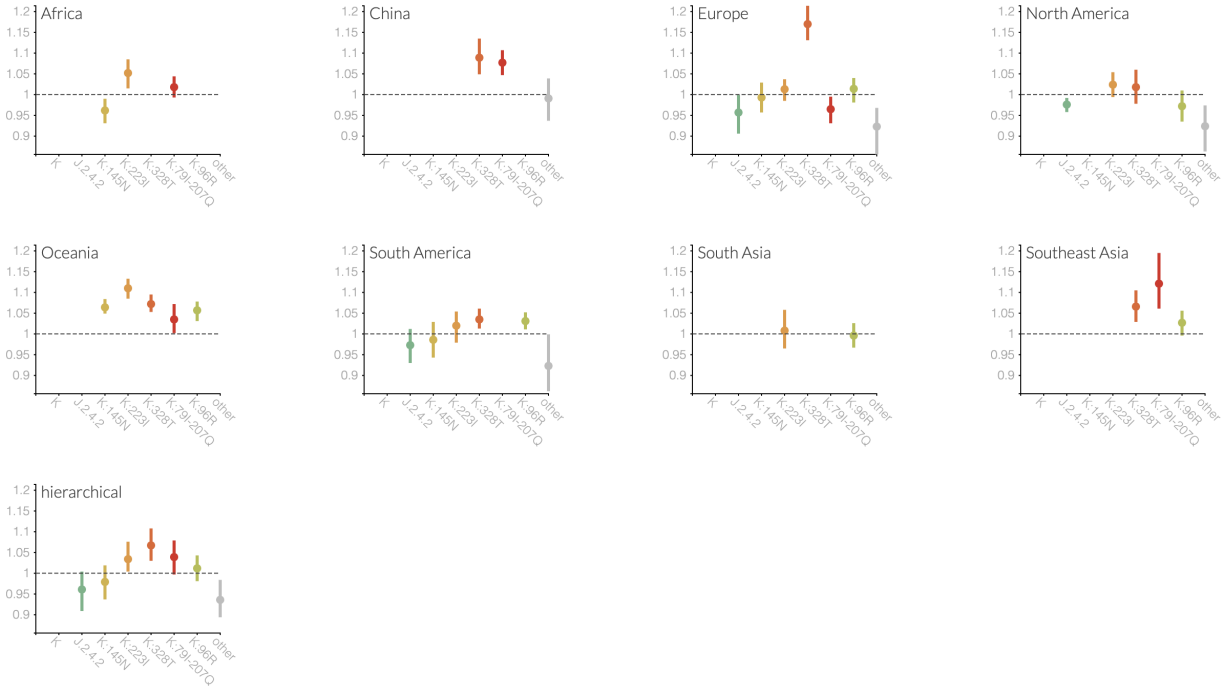


Figure 14. Growth advantages (median $\pm$ 95% HPDIs) per clade by region as estimated by a MLR model. The “hierarchical” advantage represents the global average fitness per clade. Advantages are relative to a pivot clade (K) with value 1. Inferences of growth advantages >1 indicate that a clade is more fit than the pivot clade, while values <1 indicate that a clade is less fit than the pivot clade. View on Nextstrain.

Antigenic properties from ferret serology

Synthesizing all ferret HI measurements into our titer substitution model, we find the highest antigenic advance for K viruses with HA1:135N or HA1:145R substitutions (Fig. 15). K:145N viruses appear to be the largest group that is the most antigenically advanced by ferret sera.

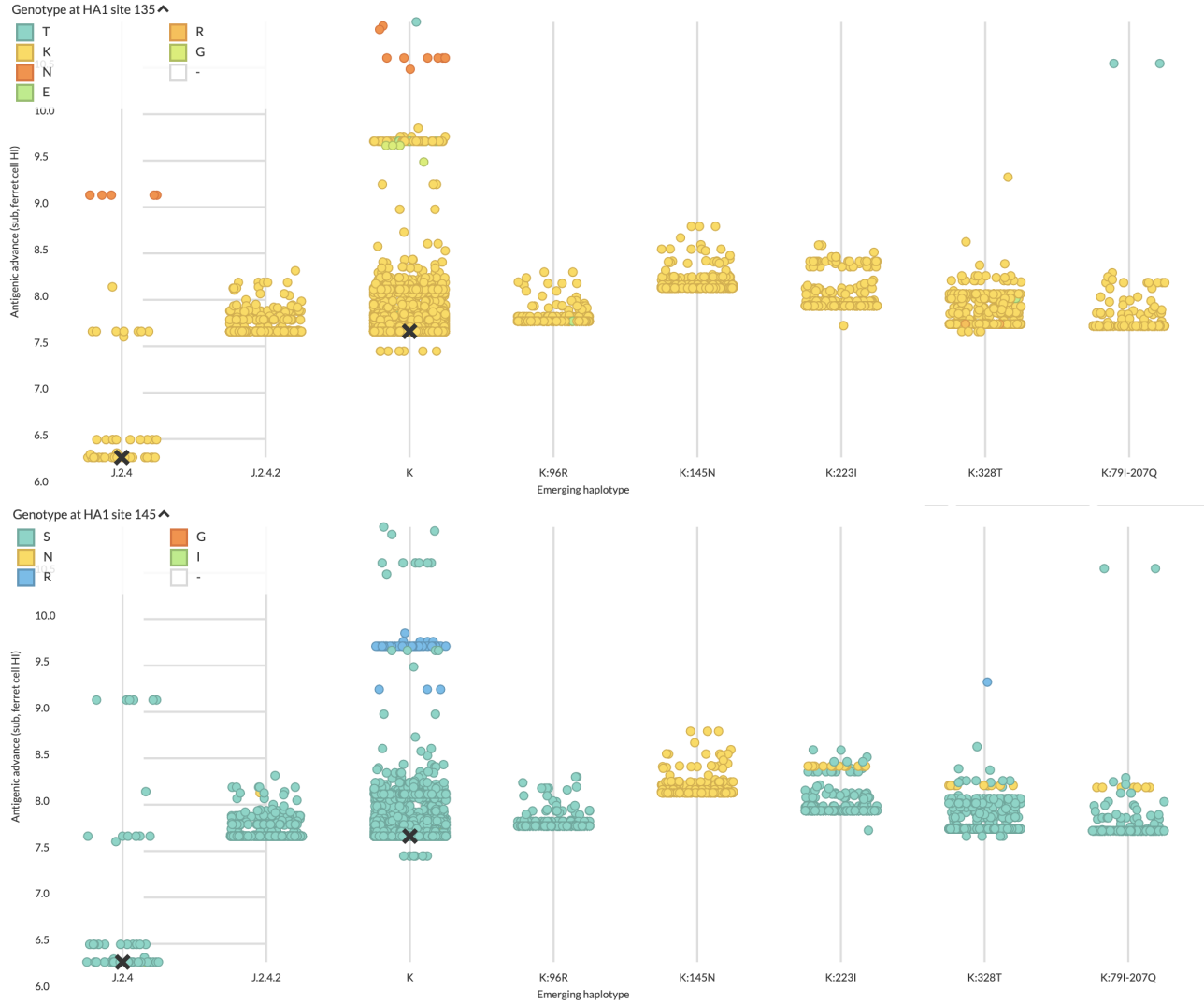


Figure 15. Antigenic advance per clade estimated from ferret-based HI data [7] for samples collected since February 1, 2026. In each panel, points represent individual HA sequences whose overall antigenic advance has been inferred from a titer model based on the corresponding raw serological data. Points are colored by their genotype at position HA1 135 to highlight the antigenic advance of J.2.4 strains with HA1:135N. View on nextstrain.org.

[Raw ferret data redacted]

Antigenic properties from human serology

The Bloom Lab measured neutralization titers for recent H3N2 and H1N1pdm strains against human sera collected in April through August 2026 (see methods summary at the beginning of the report). Results and data are available in a preprint posted on bioRxiv [8] and summarized in the figures below (Fig. 16). Note that viruses from K with the HA:145R substitution did not grow in laboratory conditions for the neutralization assays. This variant is not represented by the human data, although it shows strong antigenic advance in ferrets.

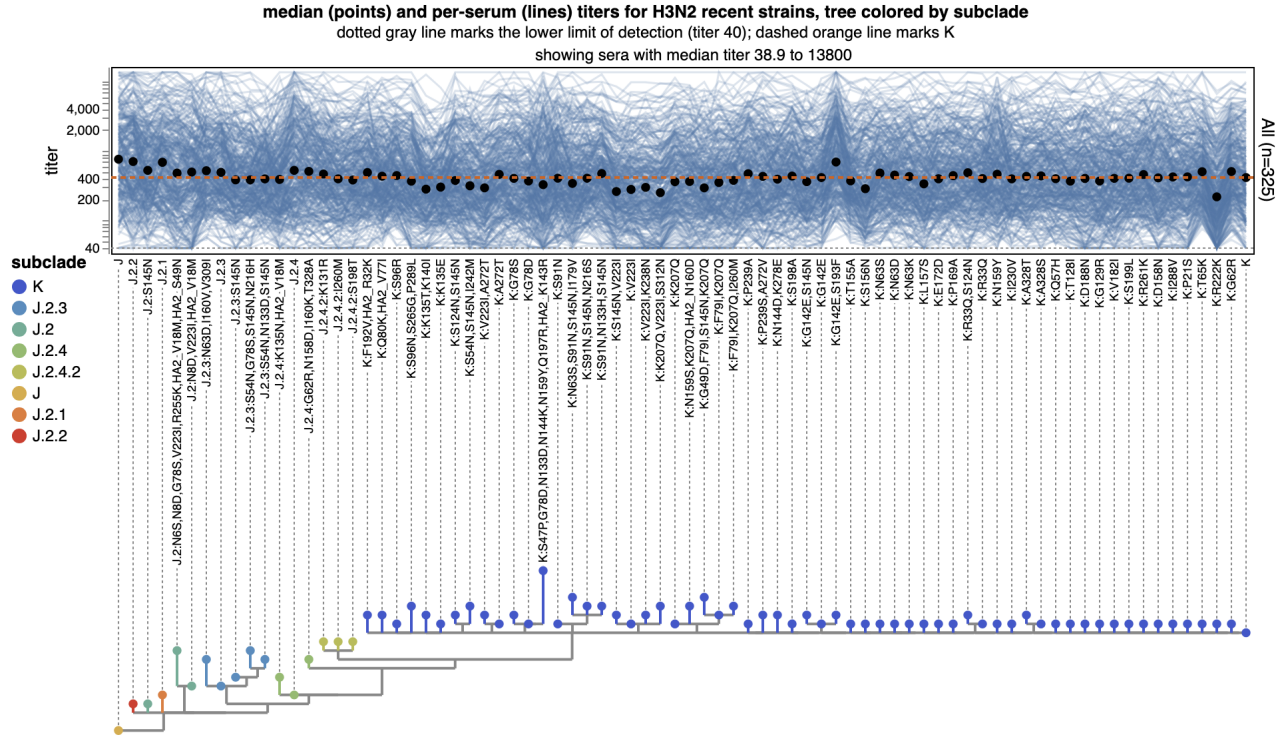


Figure 16. Tree summarizing H3N2 strains used by Kikawa, Butler et al. [8] and individual titers per virus and serum. The solid points represent the median titer across all sera. The dashed line represents the titer to the current H3N2 vaccine strain.

Titer models fit to these neutralization titers show the highest antigenic advance K viruses with 222K or 223I substitutions (Fig. 17).

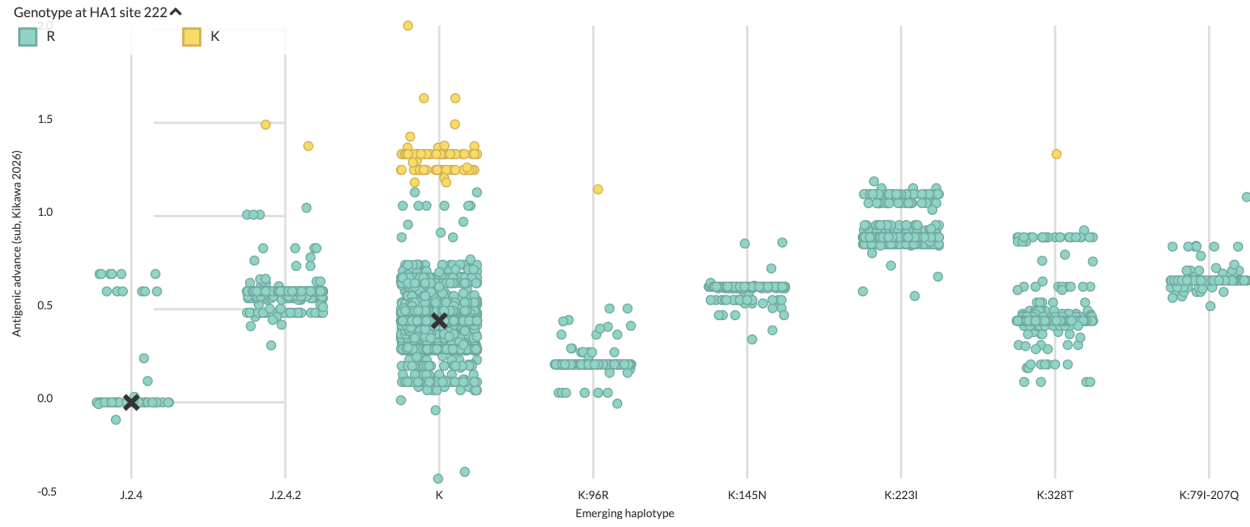


Figure 17. Antigenic advance across clades estimated from human-based neutralization data [10] for samples collected since May 1, 2025. Points are colored by their genotype at HA1 222. View on nextstrain.org.

Comparing viral fitness forecasts and antigenic phenotype

To understand the relationship between viral fitness estimated from sequence data alone (growth advantages from the MLR model) and antigenic advance from neutralization assays, we plotted the growth advantage of each recent emerging haplotype against the corresponding median antigenic advance of sequences in that haplotype based on human or ferret serological data. Haplotypes K:223I and K:79I-209Q show the most antigenic advance for humans which corresponds to the high fitness of these variants (Fig. 18). In contrast, K:96R has slightly higher fitness than K but the lowest antigenic advance. K:328T does not differ from K antigenically. Since 328T has been the dominant allele for the entire history of H3N2 until 328A swept as part of K, we hypothesize that 328A is a lower fitness substitution that hitchhiked with K's other high fitness substitutions. K:145N is most antigenically advanced based on ferret sera, while this haplotype has a lower fitness than K. Other H3N2 haplotypes show a strong positive correlation between antigenic advance from ferrets and fitness.

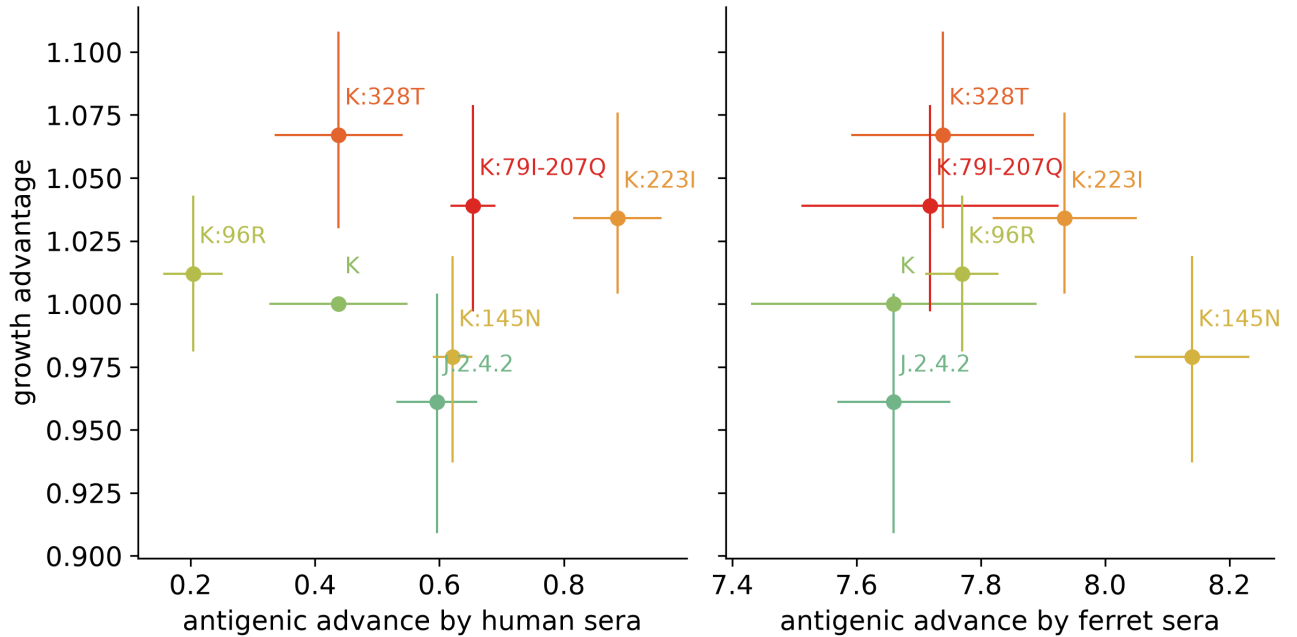


Figure 18. Growth advantage by antigenic advance per clade estimated with a titer substitution model [7] for human-based neutralization data [10] (left) and ferret-based cell HI data (right). Growth advantages shown as median $\pm$ 95% HPDIs. Antigenic advance shown as median $\pm$ 1 standard deviation across all sequences in a given clade.

Finally, we ranked the available vaccine candidates by their antigenic distance to the predicted future H3N2 population as described in the Methods. Results from both human and ferret antigenic models found that K reference viruses like A/Darwin/1415/2025 were closest to the predicted future (Figs. 19 and 20). Based on human serology, the K:79I-207Q reference virus was the closest to the predicted future with a distance of 0.11 $\log_2$ units. The A/Darwin/1415/2025 vaccine was 0.17 $\log_2$ units from the predicted future, indicating that it is still a good fit.

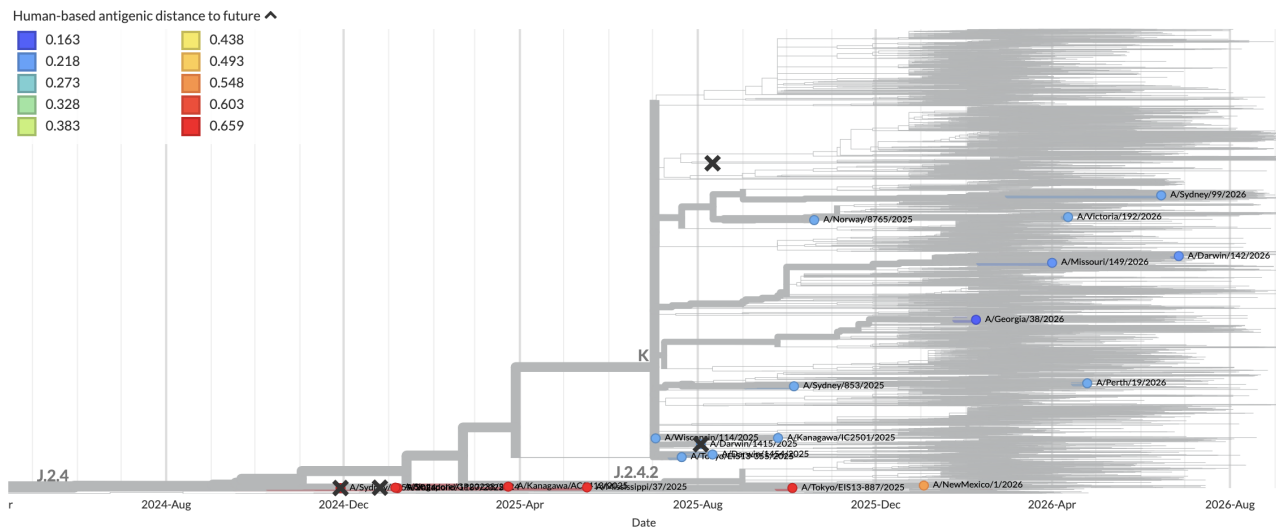


Figure 19. Weighted average antigenic distance to the future population per reference virus using human-based neutralization data. For each reference virus, we calculated the average antigenic distance between that virus's HA1 amino acid sequence and the corresponding HA1 sequences for viruses from each emerging haplotype and geographic region. The pairwise antigenic distance between viruses was based on the HA1 substitutions between them and antigenic weights per substitution estimated with a titer substitution model [7] fit to human-based neutralization data [10]. View on nextstrain.org.

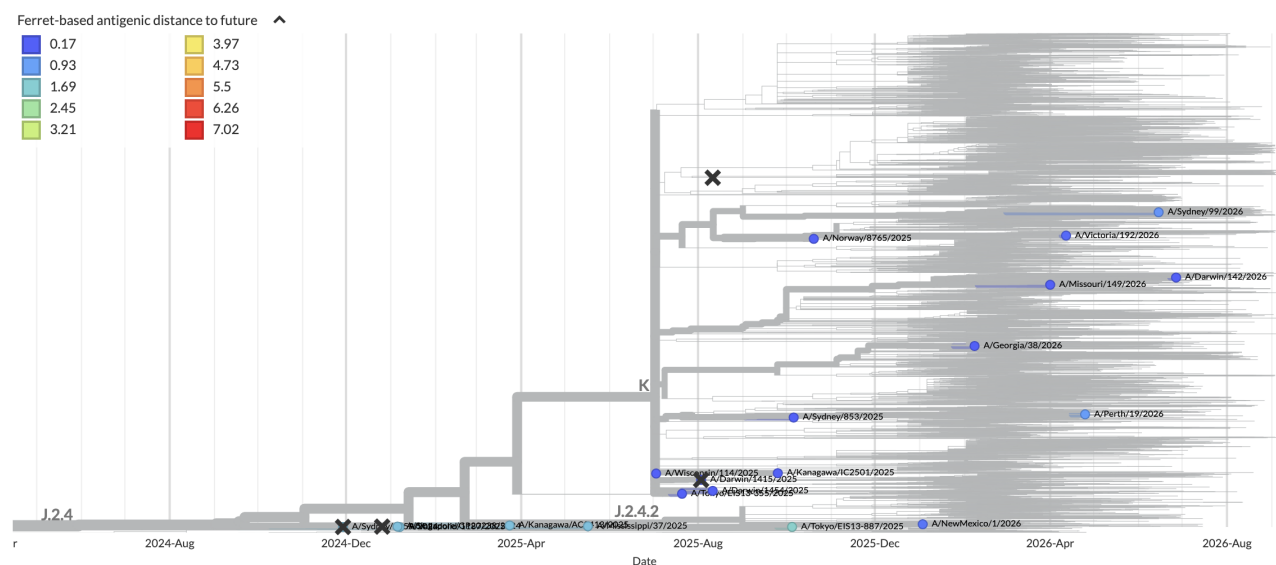


Figure 20. Weighted average antigenic distance to the future population per reference virus using ferret-based cell HI data. For each reference virus, we calculated the average antigenic distance between that virus's HA1 amino acid sequence and the corresponding HA1 sequences for viruses from each emerging haplotype and geographic region. The pairwise antigenic distance between viruses was based on the HA1 substitutions between them and antigenic weights per substitution estimated with a titer substitution model [7] fit to ferret-based cell HI data. View on nextstrain.org.

B/Vic

Summary

B/Victoria HA diversity continues to persist with C.3 and C.5 descendants co-circulating and diversifying (Fig. 21). Although clade composition differs across regions, C.3.1, C.3.3, C.5.1, and C.5.7 all appear to be declining in frequency globally (Fig. 22). C.5.6.1:241Q is growing in multiple regions followed closely by C.5.6. C.5.6.1:241Q has a slightly higher fitness than C.5.6 and C.5.6.1 in the regions where both have circulated (Fig. 23). Beyond these well-established variants, a C.5.6 subclade with HA1:E183K is growing in Europe and Oceania and has circulated this year in every region.

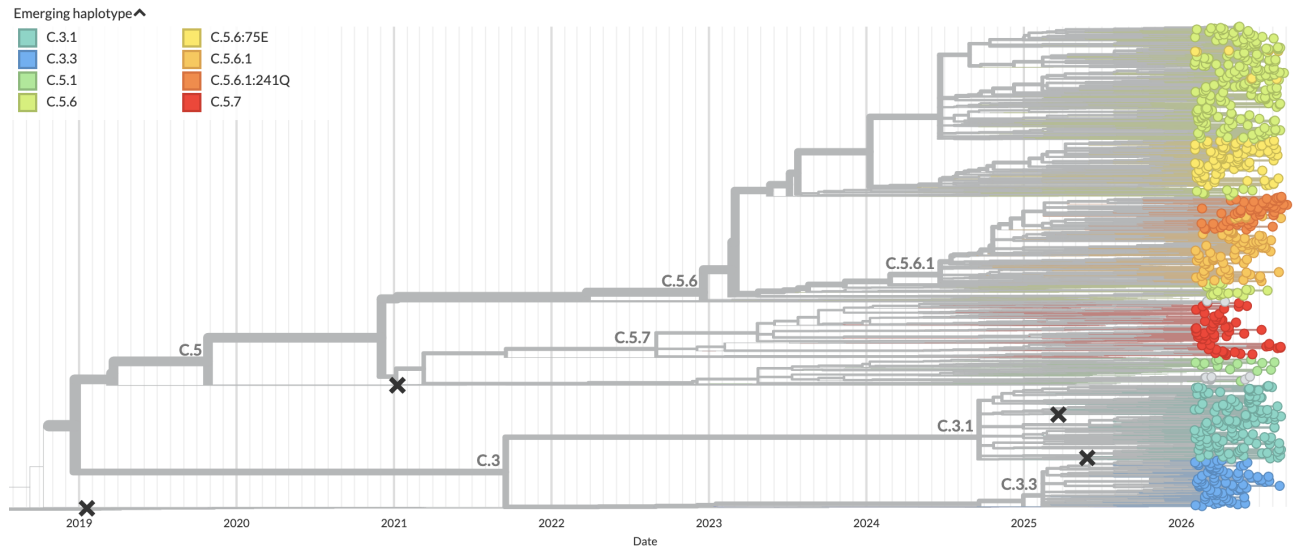


Figure 21. Time-resolved B/Victoria phylogeny colored by clade and filtered to strains collected since February 1, 2026. View on nextstrain.org.

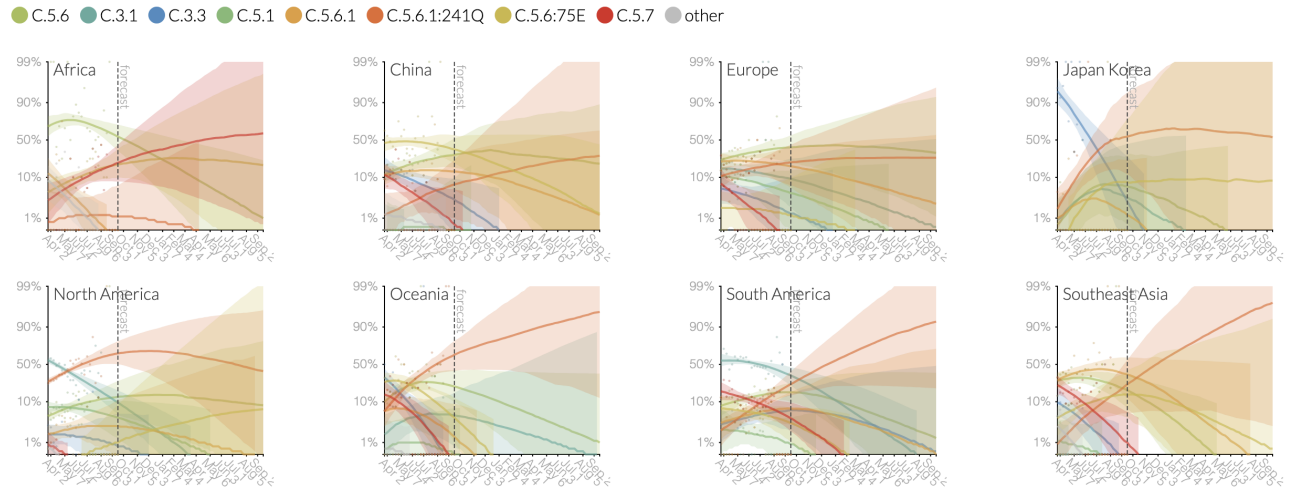


Figure 22. Clade frequencies by region estimated by a multinomial logistic regression (MLR). Lines show the median frequency ($\pm$ 95% HPDIs) and 1-year forecasts. Colored points indicate frequency estimates from the raw data. View on Nextstrain.

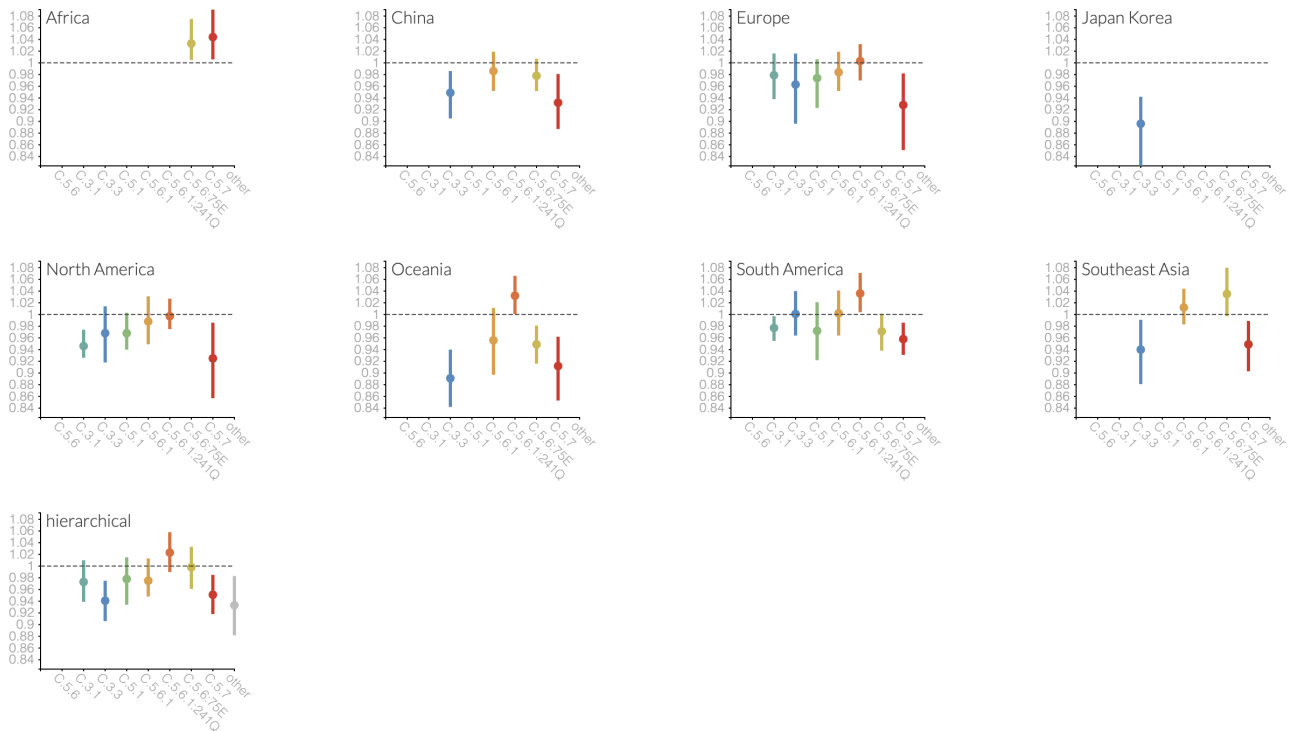


Figure 23. Growth advantages (median $\pm$ 95% HPDIs) per clade by region as estimated by a MLR model. The “hierarchical” advantage represents the global average fitness per clade. Growth advantages are relative to a pivot clade (C.5.6) at 1. View on Nextstrain.

Antigenic properties from ferret serology

C.3 descendants with the HA1:197N substitution have the highest antigenic advance of any clade by ferret HI (Fig. 24). Within C.5 clades, HA1:241K and 241Q substitutions increase antigenic advance by 0.76 and 0.35 $\log_2$ units, respectively.

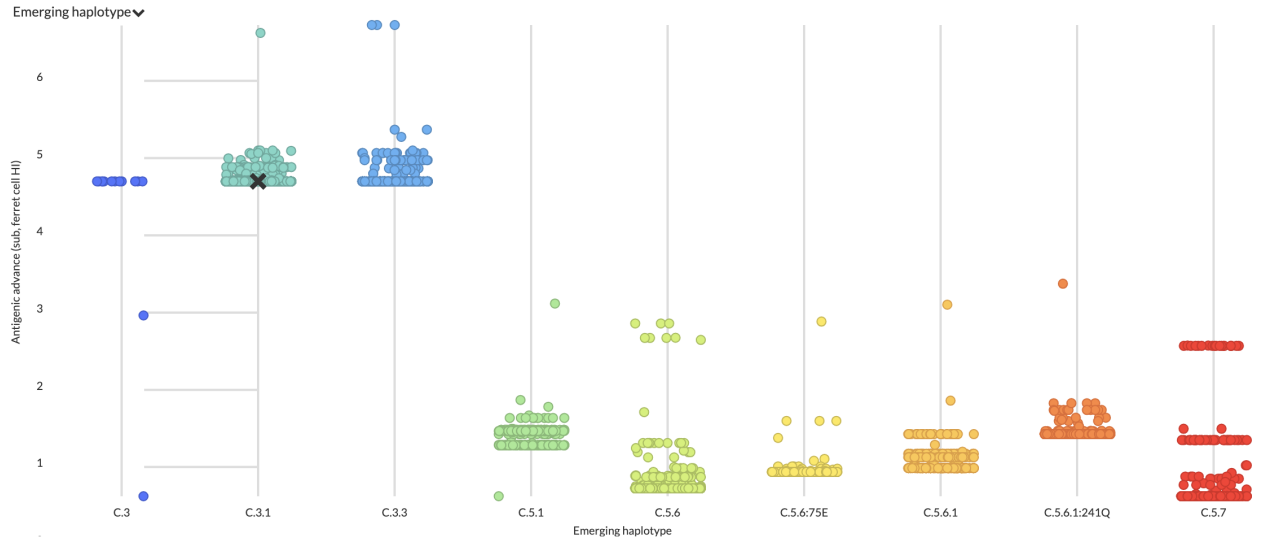


Figure 24. Antigenic advance per clade estimated from ferret-based HI data [7] for samples collected since February 1, 2026. View on nextstrain.org.

[Raw ferret data redacted]

Acknowledgments

This work is made possible by the timely sharing of influenza virus sequence data through GISAID. A table acknowledging each individual data contributor to the analysis on nextstrain.org/seasonal-flu can be found online at nextstrain.org/seasonal-flu. We thank the Influenza Division at the US Centers for Disease Control and Prevention, the Victorian Infectious Diseases Reference Laboratory at the Australian Peter Doherty Institute for Infection and Immunity, the Influenza Virus Research Center at the Japan National Institute of Infectious Diseases, the Crick Worldwide Influenza Centre at the UK Francis Crick Institute for data sharing and feedback. We thank Rebecca Kondor and the CDC Influenza Division team for insight regarding analysis directions.

References

1. Neher RA, Bedford T (2015) nextflu: real-time tracking of seasonal influenza virus evolution in humans. *Bioinformatics* 31: 3546–3548.
2. Hadfield J, Megill C, Bell SM, Huddleston J, Potter B, et al. (2018) Nextstrain: real-time tracking of pathogen evolution. *Bioinformatics* 34: 4121–4123.
3. Shu Y, McCauley J (2017) Gisaids: Global initiative on sharing all influenza data – from vision to reality. *Eurosurveillance* 22.
4. Ly-Trong N, Bielow C, De Maio N, Minh BQ (2024) CMAPLE: Efficient Phylogenetic Inference in the Pandemic Era. *Molecular Biology and Evolution* 41.
5. Aksamentov I, Roemer C, Hodcroft EB, Neher RA (2021) Nextclade: clade assignment, mutation calling and quality control for viral genomes. *Journal of Open Source Software* 6: 3773.
6. Abousamra E, Figgins M, Bedford T (2024) Fitness models provide accurate short-term forecasts of SARS-CoV-2 variant frequency. *PLoS Comput Biol* 20: e1012443.
7. Neher RA, Bedford T, Daniels RS, Russell CA, Shraiman BI (2016) Prediction, dynamics, and visualization of antigenic phenotypes of seasonal influenza viruses. *Proc Natl Acad Sci USA* 113: E1701–E1709.
8. Kikawa C, Butler A, Huddleston J, Turner SA, Peck H, et al. (2026) Near real-time data on the human neutralizing antibody landscape to influenza virus in summer of 2026 shows antigenic advance of H3N2 subclade K region D mutants and H1N1 D.3.1.1 Sa mutants. *bioRxiv* .
9. Kikawa C, Huddleston J, Loes AN, Turner SA, Lee J, et al. (2025) Near real-time data on the human neutralizing antibody landscape to influenza virus to inform vaccine-strain selection in September 2025. *bioRxiv* : 2025.09.06.674661.
10. Kikawa C, Huddleston J, Turner SA, Loes AN, Liu J, et al. (2026) Near real-time data on the human neutralizing antibody landscape to influenza virus as of early 2026 to inform vaccine-strain selection. *bioRxiv* .